arXiv Corpus Statistics Report

Project: arXiv Hybrid Information Retrieval System
Category: cs.CV
Date Range: 2023-01-01 to 2026-10-31

1. Corpus Overview

Metric	Value
Total unique papers	593
Number of subtopics	4
Successful text extractions	593
Failed text extractions	0
Missing titles	0
Missing abstracts	0
Average pages per paper	17.07
Average extracted characters	58,649.53

2. Subtopic Distribution

Subtopic	Number of Papers
Object Detection	150
Pose Estimation	149
Image Segmentation	148
Medical Imaging	146

3. Paper Length Distribution

Length Category	Definition	Number of Papers
Short	Less than 6 pages	28
Medium	7 to 9 pages	81
Long	More than 9 pages	460
Unclassified	Exactly 6 pages	24

Note: The assignment defines short papers as fewer than 6 pages, medium papers as 7–9 pages, and long papers as more than 9 pages. Papers with exactly 6 pages do not fall into any of these specified categories, so they are reported separately as unclassified.

4. Data Collection and Processing

The corpus was collected from the arXiv cs.CV category within the specified date range. It contains four subtopics: object detection, pose estimation, image segmentation, and medical imaging. Paper metadata and extracted text are stored locally.

5. Retrieval System

The project implements BM25 sparse retrieval and dense retrieval using sentence embeddings. Their rankings are combined using Reciprocal Rank Fusion (RRF) with k = 60.